

Adaptive and Distributed Radio Resource Allocation in Densely Deployed Wireless LANs: A Game-Theoretic Approach

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Abstract—As increasing numbers of access points (APs) are deployed in wireless local area networks (WLANs), they experience severe channel interference. Thus, a well-defined resource management scheme for APs should be devised. Undoubtedly, a centralized and deterministic resource management is not appropriate, owing to the distributed operation and dynamic channel condition of densely deployed WLANs. To address this problem, an adaptive and distributed radio resource allocation (ADRA) algorithm is proposed, in which an optimal mixed strategy for selecting the channel width of APs is derived via a game-theoretic approach. ADRA also fits into the densely deployed WLAN in that it considers partially overlapped channels. Simulation results show that ADRA can provide a comparable throughput and fairness performance to centralized algorithms, even though it functions in a distributed manner. Moreover, it is shown that ADRA works adaptively even when network environments change rapidly.

Index Terms—Adaptive algorithm, resource allocation, distributed MAC, dense WLAN, game-theory.

I. INTRODUCTION

AN INCREASING number of multimedia devices, including smartphones, laptops, and high-definition devices, bring about a need for improving the network throughput. Since these devices are mostly connected to the media server via wireless local area networks (WLANs), a staggering number of APs have been installed. As a result, there are numerous problems, such as severe signal interference and heavy congestion due to the overcrowded APs, as the primitive IEEE 802.11 standard was designed only for a single service set. Therefore, a need for an appropriate algorithm for distributing radio resources has arisen.

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Radio resource allocation (RRA) is the one of the promising technologies for mitigating co-channel interference between devices. There are various RRA techniques for WLAN that are based on the basic RRA algorithm. However, previous studies only considered centralized and static environments. That is, previous algorithms require a centralized controller connected to all APs, and the controller must be aware of the states of the APs. Because such centralized and static channel allocation cannot satisfy the dynamic and fluctuated traffic demands of APs [2], the development of an adaptive and distributed RRA algorithm is inevitable.

In this paper, an adaptive and distributed RRA (ADRA) algorithm is proposed with consideration of partially overlapped channels in densely deployed wireless LANs. We focus on how to allocate the channel bandwidth to the associated WLAN APs via a game-theoretic approach. ADRA achieves an improved the network throughput as well as fairness among the APs. ADRA can handle dynamic environments because an accidental case such as the appearance or disappearance of new APs or changing their channel widths can be reflected by means of the periodically computed utility function. Moreover, because the AP does not need information about other APs, except for the observed throughputs, ADRA can operate in a distributed manner and does not require centralized processing. Extensive simulation results demonstrate that ADRA has comparable performance to centralized algorithms, i.e., the auto-configuration framework for enterprise 802.11n WLANs (ACORN) [3] and the proportional fairness (PF) channel allocation algorithm [4].

The main contributions of ADRA can be summarized as follows. 1) ADRA is adaptive and thus can handle dynamic situations such as the fluctuation of the wireless channel and the registration or leaving of APs. That is, any sudden change in the wireless environment can be handled. 2) ADRA is flexible, allowing it to conduct preferential treatment between the network throughput and the fairness among APs by adjusting a weighting factor. 3) ADRA considers partially overlapped channels; thus, its throughput can be further improved in densely deployed networks.

The rest of the paper is organized as follows. First, a brief review of related works is provided in Section II. The system model for the ADRA algorithm is described in Section III. The resource-allocation algorithm is described in Section IV, and the impact of our algorithm is analyzed in Section V. Section VI concludes the paper.

II. RELATED WORKS

There have been numerous studies on resource allocation/distribution in cellular networks [5]–[7], whereas the literature on resource distribution, which is called channel bonding, based on IEEE 802.11n is scarce because it only supports 20 or 40 MHz. Arslan *et al.* conducted an experimental study to understand the implications of wide channels on the throughput [3]. Specifically, it was shown that the wide channel has disadvantages and may degrade the performance for IEEE 802.11n. Deek *et al.* identified network factors that influence the performance of channel bonding in IEEE 802.11n and made intelligent channel bonding decisions [8]. Lee *et al.* compared two multi-channel access schemes based on IEEE 802.11n: 1) channel bonding and 2) concurrently accessible medium access control (MAC) with multiple transceivers [9]. Unfortunately, because there are only two selectable choices in IEEE 802.11n, few studies on resource allocation for IEEE 802.11n have been conducted.

Regarding IEEE 802.11ac and subsequent standards, Bellalta *et al.* analyzed the interaction between a group of neighboring APs and evaluated the impact on the achievable throughput using a continuous-time Markov network model [4]. Joshi *et al.* proposed an analytical model for investigating the average channel throughput at the MAC layer according to Markov chain analysis [10]. Herzen *et al.* designed an algorithm for jointly assigning both the center frequency and the channel width [11]. The proposed algorithm requires out-of-band measurements in order to make spectrum allocation decisions. Yuan *et al.* designed a distributed algorithm for achieving Nash equilibrium (NE) channel-width allocation that globally maximizes the network utility and proposed a punishment-based self-enforcement mechanism to ensure that the APs obey the proposed algorithm [12]. Wang *et al.* proposed a channel bonding algorithm based on adaptive channel clear assessment (CCA) [13]. In this algorithm, each node adjusts its CCA threshold according to the link quality and interference conditions to select an appropriate bandwidth.

Other approaches have been proposed for utilizing limited resources by splitting channel bandwidths. Tan *et al.* argued that the channel should be divided into separate subchannels and introduced a fine-grained channel access method [14]. Furthermore, Yun *et al.* developed a spectrum adaptation prototype utilizing a fine-grained channel access method [15].

Supporting a partially overlapped channel assignment is the one of the most promising techniques for dense networks. Rayanchu *et al.* introduced a system model for designing WLANs using flexible channelization, which determines an appropriate channel width and a center frequency for each transmission for IEEE 802.11n [16]. Mishra *et al.* revealed that a partially overlapped channel may not be harmful [17]. They modeled partially overlapped channels and confirmed that the careful use of partially overlapped channels can often lead to a significant improvement in the spectrum utilization and application performance. Cui *et al.* derived an interference model that considers both the adjacent channel separation and the physical distance of the two nodes employing adjacent channels [18]. Duarte *et al.* explored the possibility of exploiting partially

overlapped channels by introducing a novel game-theoretic distributed channel-assignment algorithm [19]. Zhou *et al.* studied how to reduce the interference and improve the network throughput by adopting partially overlapped channels in the signal-to-interference-plus-noise ratio (SINR) model [20]. Xu *et al.* investigated the problem of distributed channel selection in a network with partially overlapped channels using a game-theoretic learning algorithm [21], [22].

Furthermore, there have been several studies focusing on the efficient utilization of the limited bandwidth in wireless networks. Wang *et al.* suggested a frame-level wideband spectrum adaptation scheme [23]. Specifically, they introduced a baseband processing layer between the PHY/MAC layer and the radio frequency (RF) front-end and proposed a wideband-detectable preamble structure to indicate the new transmission band. Additionally, Wang *et al.* proposed non-cooperative and cooperative joining strategies for cognitive radio networks [24]. Using a game-theoretic approach, they derived the NE and optimal joining strategies for the case where secondary users (SUs) are non-cooperative or cooperative. Then, to bridge the gap between the individually and globally optimal strategies, they proposed an admission fee for SUs to join the bandwidth.

Kai *et al.* suggested an optimal channel allocation algorithm [25]. They adopted a continuous-time Markov Chain (CTMC) model to analyze the equilibrium throughput. Based on the analysis, they formulated an integer nonlinear programming (INLP) model with a target of maximizing the system throughput. Additionally, Khairy *et al.* proposed an analytical framework that allows contiguous channels whose length is a power of two and presented detailed performance evaluation results [26]. Faridi *et al.* analyzed the opportunistic multi-channel bonding mode in IEEE 802.11ac WLANs with co-existing legacy IEEE 802.11a/b/g users by means of a two-level renewal process [27]. Although they investigated densely deployed wireless networks, they did not consider partially overlapped channels and fairness between APs.

III. SYSTEM MODEL

Fig. 1 shows an illustrative topology in which a group of single-hop basic service sets (BSSs) is assumed. Each BSS consists of one AP and one or more devices. Throughout this paper, IEEE 802.11ac is assumed, but our algorithm can be adapted to any IEEE 802.11 standard.

In our model, each AP is a rational player in the proposed game model, which tries to determine its channel width to maximize its utility. We refer to a certain AP (e.g., AP k in Fig. 1) as a row player and the adjacent APs within the transmission range of the row player (e.g., J_k in Fig. 1) as column players. That is, APs adjacent to a certain AP are grouped together, and the group is referred to as the column player from the viewpoint of the row player. As a result, the number of games equals the number of BSSs in which there are row and column players. The row player calculates its gain and loss functions via ADRA and derives a mixed strategy profile from the utility obtained from the gain and loss functions.

ADRA considers partially overlapped channels; thus, a model for partially overlapped channels should be considered. The

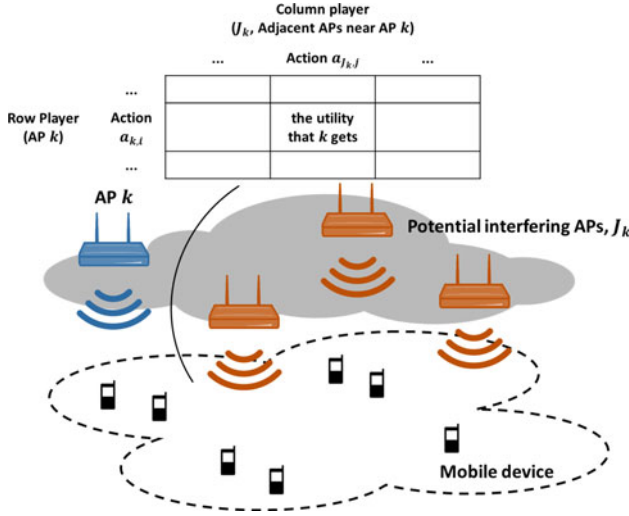
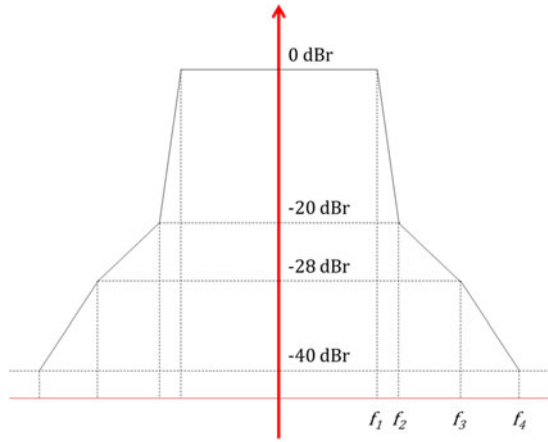


Fig. 1. An illustrative topology.



Channel Size	f_1	f_2	f_3	f_4
20 MHz	9 MHz	11 MHz	20 MHz	30 MHz
40 MHz	19 MHz	21 MHz	40 MHz	60 MHz
80 MHz	39 MHz	41 MHz	80 MHz	120 MHz
160 MHz	79 MHz	81 MHz	160 MHz	240 MHz

Fig. 2. Spectral mask for the IEEE 802.11ac.

notion of an interference factor (I-factor) was introduced to determine the amount of overlap between two channels [17]. Let P_i and P_j denote the received power from a particular signal on channels i and j , respectively. Then, the amount of overlap from the viewpoint of channel i caused by the signal from channel j , which is denoted by $I(i, j)$, and can be defined as P_i/P_j . This indicates that the I-factor can be obtained through simple measurements. For example, using the pre-defined spectral masks from the IEEE 802.11 standard as illustrated in Fig. 2, the I-factor of any two channels for any channel widths and center frequencies can be calculated.

IV. ADAPTIVE AND DISTRIBUTED RESOURCE ALLOCATION ALGORITHM

First, the utility calculation module tries to construct a utility matrix that is used to develop the optimal mixed strategy. This utility matrix consists of two functions: the gain and

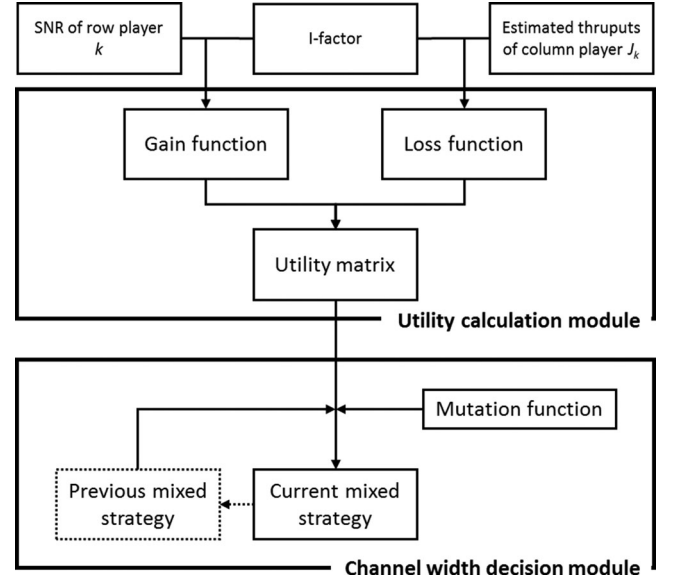


Fig. 3. High level of modules and functions of ADRA.

loss functions. The gain function represents the attainable throughput when a certain player k selects a pure strategy, while the loss function refers to the opportunity cost of corresponding players to player k . The balance between the fairness of the players and overall throughput can be prioritized by adjusting the parameter α .

Next, the bandwidth decision module exploits the utility matrix to calculate the mixed strategy. With a multiplicative weights method [28], the mixed strategy of each player is derived from the utility matrix. Each player picks its action from the strategy set according to the mixed strategy and the previous mixed strategy. These results can be reset by a mutation function; thus, ADRA can be adaptive even in dynamic and fluctuating situations, such as the registration or leaving of APs. Above-mentioned modules and functions are illustrated in Fig. 3 and key notations in ADRA are summarized in Table I.

A. Utility Calculation Module

Because there are $|K|$ players, $|K|$ matrices can be defined in ADRA. In each matrix, the row player represents a particular AP, and the column players represent one or more adjacent APs near the selected AP.

Let M_k be a matrix with $|A_k|$ rows and $|A_{J_k}|$ columns for row player k . The rows of M_k are the pure strategies of row player k , and the columns are a combination of the pure strategies of column player J_k , which is a group of APs near player k . The j th entry of the i th row of the matrix M_k is defined as $M_k[i, j]$, which represents the utility when the row player k performs the i th pure action and the column player of player k performs the j th pure action. That is, $M_k[i, j] = u_k(a_{k,i}, a_{J_k,j})$, where $u_k(a_{k,i}, a_{J_k,j})$ represents the utility of player k , as shown in Fig. 4.

The utility consists of the gain and loss functions. The obtainable throughput in the case where row player k selects a certain channel width is included in the gain function— $u_{k,G}(a_{k,i}, a_{J_k,j})$ —while the sum of the opportunity costs of

TABLE I
SUMMARY OF NOTATIONS

Modeling parameter	Definition
K	The set of players.
$A_k = \{a_{k,1}, \dots, a_{k,N}\}$	The strategy space of the row player k . Player k has N pure strategies in his strategy space.
$A_{J_k} = \{a_{J_k,1}, \dots, a_{J_k,M}\}$	The strategy space of the column player J_k . M is the product of the number of AP's pure strategies in J_k .
$q_{k,t}$	The mixed strategy for the row player k at time t . It is a probability distribution over the N pure strategies in player k 's strategy set.
$q_{J_k,t}$	The mixed strategy for the column player J_k at time t . It is a probability distribution over the M pure strategies in player J_k 's strategy set.
M_k	The matrix with the utility that the row player k gets.
$M_k[i, j] = u_k(a_{k,i}, a_{J_k,j})$	The utility when the row player k acts the i th pure strategy and the column player J_k does the j th pure strategy.
$u_{k,G}(a_{k,i}, a_{J_k,j})$ and $u_{k,L}(a_{k,i}, a_{J_k,j})$	The gain and the loss functions when the row player k acts the i th pure strategy and the column player the J_k does j th pure strategy.
$U_k(q_{k,t}, q_{J_k,t})$	The expected utility.

Potential interfering players, J_1			
	...	Action $a_{J_1,j}$	...
Player, 1	...		
	Action $a_{1,i}$	$M_1[i, j]$ $= u_1(a_{1,i}, a_{J_1,j})$	
	...		
⋮			
Potential interfering players, J_k			
	...	Action $a_{J_k,j}$	...
Player, k	...		
	Action $a_{k,i}$	$M_k[i, j]$ $= u_k(a_{k,i}, a_{J_k,j})$	
	...		

Fig. 4. Matrix M_k explanation.

interfering APs (i.e., column players J_k) is considered in the loss function: $u_{k,L}(a_{k,i}, a_{J_k,j})$.

The gain function can be expressed as

$$u_{k,G}(a_{k,i}, a_{J_k,j}) = W(a_{k,i}) \cdot \log_2[1 + \{(1 - I_P(a_{k,i}, a_{J_k,j})) \cdot SNR(a_{k,i})\}], \quad (1)$$

where $I_P(a_{k,i}, a_{J_k,j})$ represents the I-factor for the amount of overlap between transmission on a given frequency when row player k performs the i th pure strategy and the reception at a given frequency when column player J_k performs the j th pure strategy. Because $I_P(a_{k,i}, a_{J_k,j})$ follows the form of the I-factor defined in Section III, it lies within $[0, 1]$. As mentioned in Section III, the I-factor can be estimated in advance. A simplified example is illustrated in Fig. 5, where the I-factor is calculated as the partially overlapped area divided by the channel bandwidth area when player k selects the channel according to action i .

$W(a_{k,i})$ and $SNR(a_{k,i})$ are the bandwidth in hertz and the signal-to-noise ratio (SNR), respectively, in the case where player k selects a pure strategy of $a_{k,i}$. The SNR can be measured by listening to beacon frames [3] and jointly considering the transmission bandwidth, thermal noise, and I-factor [17].

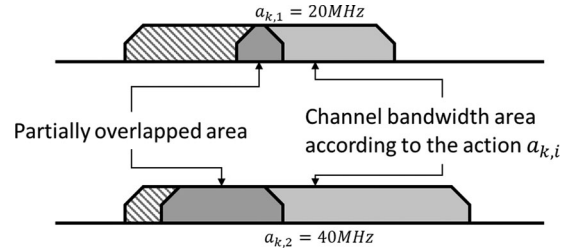


Fig. 5. An illustration for I-factor.

On the other hand, the sum of the opportunity costs when row player k selects the i th pure action $a_{k,i}$ and column player J_k selects the j th pure strategy is defined as the loss function. The loss function can be represented as

$$u_{k,L}(a_{k,i}, a_{J_k,j}) = -I_S(a_{J_k,j}, a_{k,i}) \cdot R_{J_k}, \quad (2)$$

where $I_S(a_{J_k,j}, a_{k,i})$ represents the I-factor, which is the amount of overlap between the transmission at a given frequency when column player J_k performs the j th pure strategy and reception on a certain frequency when row player k performs the i th pure strategy. Similarly, I_S also lies within $[0, 1]$.

In summary, $M_k[i, j]$ can be derived as

$$M_k[i, j] = u_k(a_{k,i}, a_{J_k,j}) = \alpha \cdot u_{k,G}(a_{k,i}, a_{J_k,j}) + (1 - \alpha) \cdot u_{k,L}(a_{k,i}, a_{J_k,j}), \quad (3)$$

where α is a weighting factor to prioritize either the attainable throughput of the player or the opportunity cost. By adjusting α , the individual throughput of a certain AP, as well as the overall network throughput, can be emphasized. For example, once α approaches to 1, the gain function is preferred, and the network throughput is prioritized. In contrast, when α approaches 0, the loss function is preferred, and the fairness is emphasized.

B. Channel Width Decision Module

After constructing the matrix M_k , mixed strategies for player k must be determined. Let $q_{k,t}$ and $q_{J_k,t}$ be the mixed strategies for row player k and column player J_k at time t , respectively.

Algorithm 1: ADRA algorithm procedure.

Require: I-factors, $A_k \forall k \in K$,
 $SNR(a_{k,i}) \forall i \in N, \forall k \in K, R_{J_k} \forall k \in K, \alpha, \beta, \gamma$.

- 1: Calculate the gain and the loss functions,
 $u_{k,G}(a_{k,i}, a_{J_k,j})$ and $u_{k,L}(a_{k,i}, a_{J_k,j}), \forall k, i, j$.
- 2: Calculate the matrix,
 $M_k[i, j] =$
 $\alpha \cdot u_{k,G}(a_{k,i}, a_{J_k,j}) + (1 - \alpha) \cdot u_{k,L}(a_{k,i}, a_{J_k,j})$
- 3: Set initial strategies, $\forall k, q_{k,0}$
- 4: **loop**
- 5: Calculate the pure strategy for selecting the i th action,
 $q_{k,t+1}(i) = q_{k,t}(i) \frac{\beta^{U_k(a_{k,i}, q_{J_k,t})}}{Z_{k,t}}$
where $Z_{k,t} = \sum_{i=1}^N q_{k,t}(i) \cdot \beta^{U_k(a_{k,i}, q_{J_k,t})}$
- 6: **if** the condition for mutation meets with respect to γ ,
then
- 7: $q_{k,t+1}$ is randomly distributed
- 8: **end if**
- 9: The player k selects $a_{k,i}$ with respect to the probability
distribution over pure strategies, $q_{k,t+1}$
- 10: **end loop**

Let the expected utility for player k be $U_k(q_{k,t}, q_{J_k,t})$ when the mixed strategy of player k is $q_{k,t}$ and the mixed strategy of the column player J_k is $q_{J_k,t}$, respectively. The expected utility $U_k(q_{k,t}, q_{J_k,t})$ can be expressed as $q_{k,t}^\top \cdot M_k \cdot q_{J_k,t}$, where $A^\top$ is the transpose of the matrix A .

The objective of ADRA is to maximize the calculated utility which is composed of the gain and loss functions, for player k . ADRA is based on the online learning approach proposed in a previous study [28]. On the t th round, a channel-width decision procedure is conducted by each player. This learning algorithm starts with an initial mixed strategy $q_{k,1}$. After the t th round, the player computes a new mixed strategy $q_{k,t+1}$ via a simple multiplicative rule as follows:

$$q_{k,t+1}(i) = q_{k,t}(i) \frac{\beta^{U_k(a_{k,i}, q_{J_k,t})}}{Z_{k,t}} \quad (4)$$

where $q_{k,t}(i)$ represents the probability that $q_{k,t}$ selects the i th pure action at the t th round. Z_t is a normalization factor, which is given by

$$Z_{k,t} = \sum_{i=1}^N q_{k,t}(i) \cdot \beta^{U_k(a_{k,i}, q_{J_k,t})}, \quad (5)$$

where $\beta \in (0, 1]$ is a convergence parameter. A low β means that the mixed strategy tends to increase or decrease faster than in the case of a high β . According to a previous study [28], the average per-trial loss, compared with the best fixed strategy, is minimized when

$$\beta = \frac{1}{1 + \sqrt{\frac{2 \ln N}{t}}}, \quad (6)$$

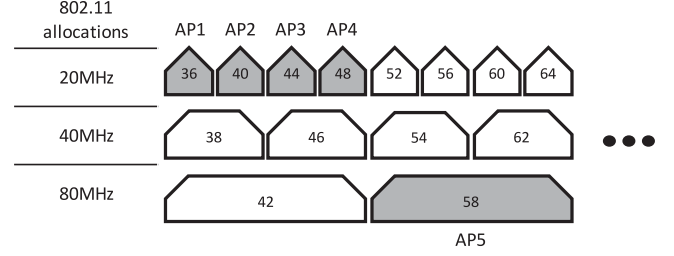


Fig. 6. Channel occupations of an operation example.

where N is the number of pure actions for the row player, and t is the number of trials. Therefore, β is fixed at this value throughout this paper.

Since the size of matrix $M_k[i, j]$ exponentially increases with the number of column players, J_k , and the number of actions, a_k, i , calculating the matrix $M_k[i, j]$ leads to high complexity. However, the complexity of ADRA is much lower than that of a centralized algorithm because the player only needs to know about its one-hop neighbor player in ADRA. Additionally, since the interval between t and $t + 1$ can be adjusted in a vendor-specific manner depending on the deployed wireless network environments, the complexity can hardly be a burden to each player.

After deriving the new mixed strategy $q_{k,t+1}$, the mutation function is applied to maintain diversity from one generation to the next. Assume that a new AP appears or disappears near a certain AP. In this case, the AP may need to change its decision. To hedge against these dynamic situations, strategies can be reset periodically or randomly. This feature can be realized via mutation, which is used in genetic algorithms, where mutation occurs during the application of the algorithm according to a pre-specified mutation probability γ . Several studies have been conducted to determine the optimal mutation probability, and [29] and [30] recommended γ values of 0.0091 and 0.011, respectively. Because the derivation of the optimal mutation probability is beyond the scope of this paper, γ is set to 0.01 in our simulations. The aforementioned procedure is summarized in Algorithm 1.

C. An Operation Example

For ease of understanding, the development of the mixed strategies of each AP is illustrated in this section. The number of players is five (i.e., $|K| = 5$), and the number of selectable pure actions is three (i.e., $N = 3$). The action space $A_k = \{a_{k,1}, a_{k,2}, a_{k,3}\}$ is given by $\{20 \text{ MHz}, 40 \text{ MHz}, 80 \text{ MHz}\}$. We set α as 0.3, and the mutation function is allowed. In this example, the basic channels of the APs are pre-defined as Ch. 36, Ch. 40, Ch. 44, and Ch. 48 for 20 MHz and Ch. 58 for 80 MHz, respectively. This setting is illustrated in Fig. 6, where the occupied channels of each AP are illustrated as a gray box. The center frequencies of AP1, AP2, AP3, and AP4 reside adjacently, while AP5 is at a relatively far-off frequency. The I-factors are also calculated according to the assigned basic channels of the APs. We assume that the initial mixed strategies are equally distributed.

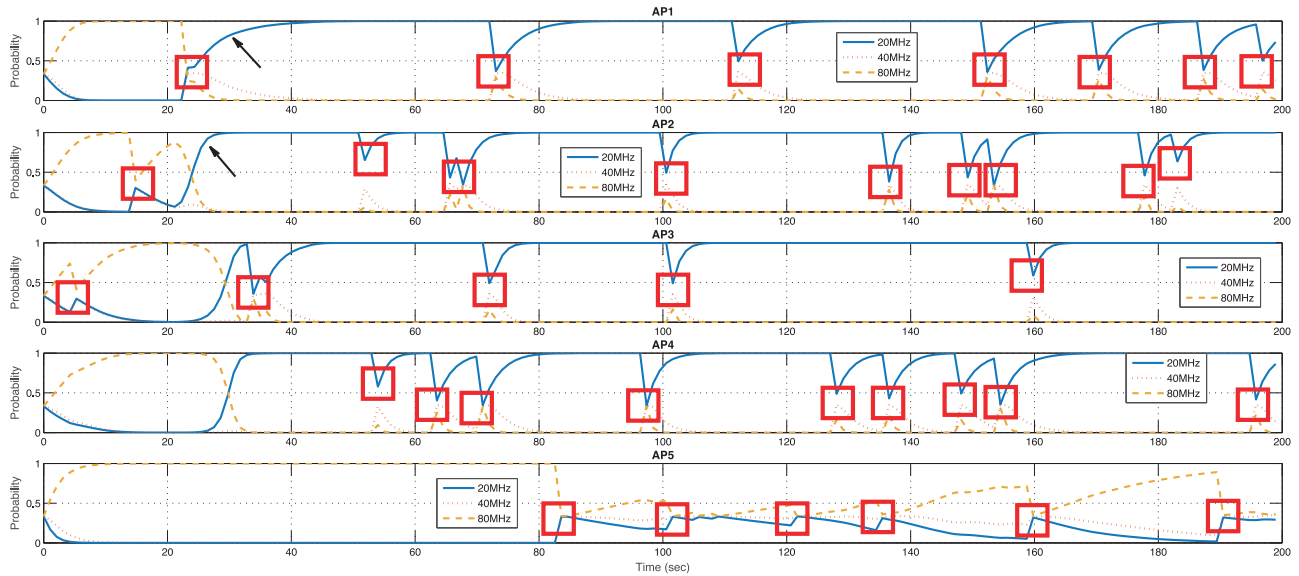


Fig. 7. Development of the mixed strategies.

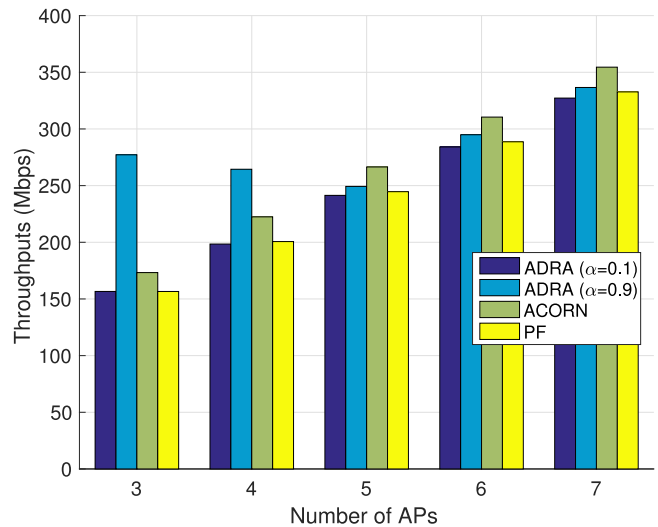
Fig. 7 shows the developments of mixed strategies according to the aforementioned scenario. Because the center frequencies of AP1, AP2, AP3, and AP4 reside adjacently, AP1, AP2, AP3, and AP4 tend to select the narrowest bandwidth: 20 MHz. On the other hand, AP5 has more opportunities to occupy wider bandwidths compared with the other APs.

Now, we examine the operations of APs from the perspective of the details of the ADRA algorithm. In Fig. 7, steep cliffs are represented by red squares. At certain times, the mixed strategy becomes neutralized by means of the invoked mutation function. The APs, except for AP5, tend to have a probability of 1 to select 20 MHz whenever the mutation function is invoked, while the probabilities of 40 and 80 MHz diminish rapidly. On the other hand, the mixed strategy of AP5 generally tends to select 80 MHz, which is the widest selectable bandwidth. This is because the center frequency of AP5 is located far from those of the other APs. As a result, the probability to select 80 MHz overwhelms the other actions (i.e., 20 and 40 MHz) whenever the mutation is invoked.

Next, we explain the effect of mixed strategies conducted by multiplicative weights. Because AP2 is surrounded by two APs while AP1 is adjacent to only by one AP, the influence of the interference caused by AP1 is far lower than that for AP2. On the other hand, the gain functions of AP1 and AP2 are exactly the same when the SNRs of the APs are the same. Hence, the utility of AP2 is lower than that of AP1. As a result, according to the multiplicative rule stated in (4) and (5), the probability of AP1 selecting 20 MHz increases faster than that of AP2 as the black arrows approach 30 s in Fig. 7. This means that the transmission bandwidth of AP2 shrinks faster than that of AP1.

V. SIMULATION RESULTS

To evaluate the performance of ADRA, extensive simulations are conducted using a MATLAB simulator. The simulation parameters are based on IEEE 802.11ac, and the results are averaged over dozens of seconds to obtain credible values.

Fig. 8. Throughputs caused by α .

The performance of ADRA is compared with 1) ACORN [3] and 2) PF [4]. In ACORN, the channel width for an AP is incremented in a greedy manner until no collisions occur owing to the overlapped channels. On the other hand, in PF, each AP doubles its channel width sequentially if available channels exist. In this manner, ACORN can maximize the total throughput while PF achieves fairness among the APs. ACORN and PF operate in a centralized manner, whereas ADRA is fully distributed. The other simulation parameters are based on the IEEE 802.11ac standard.

A. Throughputs and Fairness: ADRA Versus ACORN Versus PF

Figs. 8 and 9 shows the throughput and Jain's fairness index [31] in ADRA, ACORN, and PF. In Figs. 8 and 9, each AP is separated by up to four channels, and the SNR of AP1 is set to 20 dB, while the SNRs of the other APs are set to 15 dB.

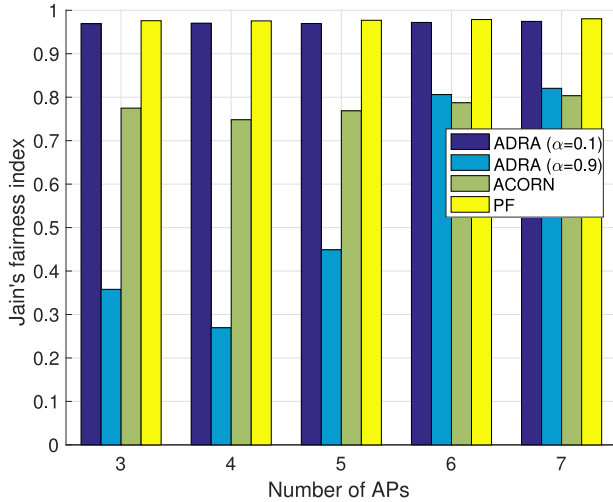


Fig. 9. Fairness indexes caused by α .

Fig. 8 shows that ACORN has the highest throughput because its goal is to maximize the throughput. ADRA with $\alpha = 0.1$ exhibits comparable throughput to PF, while ADRA and PF operate in distributed and centralized manners, respectively. This is because the strongest AP (AP1) in ADRA with $\alpha = 0.1$ prioritizes the loss function, so that the AP tends to yield the channel widths rather than occupy them. Consequently, adjacent APs may not be interrupted by the strong AP. Moreover, PF seeks to achieve the fairness among the APs and thus exhibits a similar tendency to ADRA with $\alpha = 0.1$.

Fig. 8 also demonstrates that the throughput in all schemes except ADRA, and $\alpha = 0.9$ increases with the number of APs. Interestingly, the throughput of ADRA with $\alpha = 0.9$ decreases when the number of APs increases from 3 to 5. This is because ADRA with a high α prioritizes the gain function rather than the loss function, and thus the APs in ADRA with $\alpha = 0.9$ tend to select a larger channel width compared with ADRA with $\alpha = 0.1$. Such selfish behaviors of the APs cause severe channel overlapping, which can reduce the throughput. When the number of APs exceeds 6, the effect of channel overlapping is not significant because the newly considered APs (e.g., AP6) are sufficiently separated from the first one (e.g., AP1).

Regarding fairness, Fig. 9 shows that ADRA with $\alpha = 0.1$ and PF perform comparable and high-level fairness. This is because ADRA with $\alpha = 0.1$ prioritizes the loss function, which leads to fair operation among APs. ACORN and ADRA with $\alpha = 0.9$ focus on throughput maximization instead of fair operation and thus their fairness indices are smaller than those of PF and ADRA with $\alpha = 0.1$. The fairness indices of ADRA with $\alpha = 0.9$ when the number of APs is 3 to 5 are particularly small. This is because the strongest AP (i.e., AP1) in ADRA with $\alpha = 0.9$ occupies a wider channel and therefore the transmissions of APs adjacent to AP1 are severely interrupted by AP1.

B. Effect of α

As shown in Section V-A, the performance of ADRA is greatly affected by α , and thus the effect of α is analyzed in

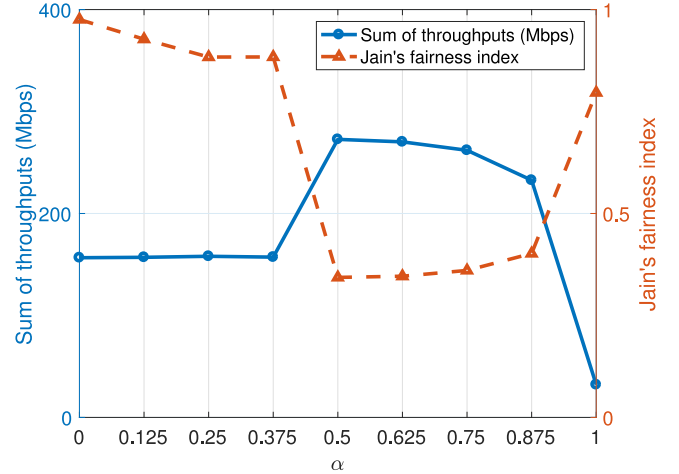


Fig. 10. Throughput and fairness index developments caused by α .

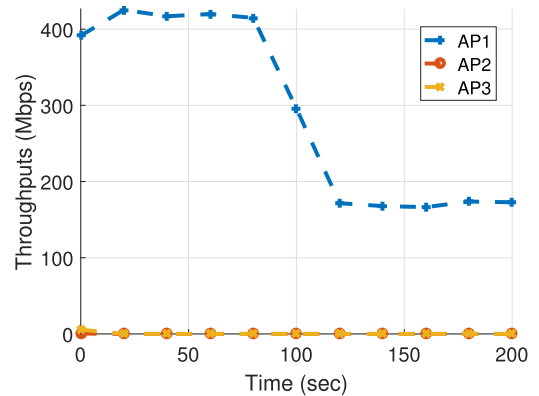


Fig. 11. Throughput development without mutation function.

Fig. 10 in more detail. The number of APs in Fig. 10 is three, and the other settings are the same as those of Figs. 8 and 9.

As shown in Fig. 10, as α increases, the throughput increases, while the fairness index generally decreases. This can be explained as follows. When α is low, the loss function is emphasized, and thus each AP sets its channel width conservatively, which leads to comparable channel widths among all the APs. On the other hand, the gain function is prioritized when α is high, and therefore each AP selects an aggressively wider channel width, which can increase its throughput. Of course, if $\alpha = 1$, only the gain function is considered in computing the utility; thus, each AP pursues its throughput maximization. Such selfish operation has a negative impact on the total throughput. As a result, it can be observed that the total throughput decreases drastically when $\alpha = 1$. Because all the APs experience degraded throughputs when $\alpha = 1$, the fairness among the degraded APs can be high (i.e., the throughput of the AP is standardized downward), as shown in Fig. 10.

C. Effect of Mutation Function

In Figs. 11 and 12, three APs are deployed. Each AP is separated by up to four channels. The SNRs of AP1 are initialized as 30 dB, whereas those of AP2 and AP3 are initialized as 15 dB.

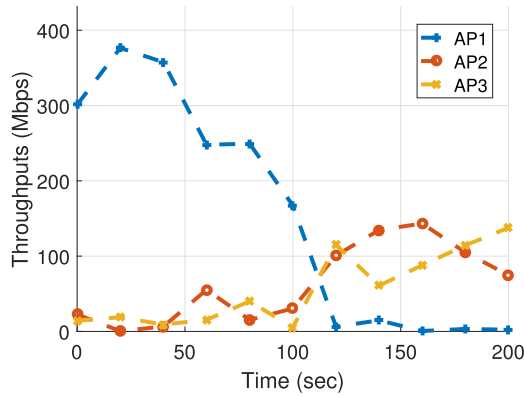


Fig. 12. Throughput development with mutation function.

SNR of AP1 decreases from 30 to 10 dB at 100 s to show the effect of the mutation function. The mutation function is not applied in Fig. 11, whereas it is applied in Fig. 12. When α is 0.9, each AP tends to prioritize its gain function rather than the loss function; thus, the APs aggressively compete for their throughputs.

As shown in Fig. 11, AP1, which has the largest SNR, can obtain a wider channel width at the beginning of the simulations. At 100 s, the SNR of AP1 is reduced from 30 to 10 dB; therefore, the throughput of AP1 decreases drastically. Nevertheless, AP1 still occupies most of the bandwidth, as AP2 and AP3 can hardly capture a sufficient channel width. This is because no mutation function is used, and AP2 and AP3 compute new strategies based on former strategies.

In Fig. 12, AP1 occupies most of the bandwidth at the beginning and experiences a drastically degraded throughput owing to the reduced SNR at 100 s. Meanwhile, AP2 and AP3 can capture their bandwidths from that time. This is because all the APs reset their mixed strategies periodically by applying the mutation function. As a result, the APs can response agilely to changes in the environment. The throughput variation in Fig. 12 is more apparent than that in Fig. 11 because the mixed strategies are periodically initialized by the mutation function.

D. Effect of Malicious Behavior

The optimized utility can be achieved only when every AP is cooperative. If a malicious AP exists, the overall utility can be reduced. Malicious behaviours can be categorized into two cases: 1) when an AP does not follow the game and 2) when an AP tries to achieve its throughput. These two cases can be regarded as that the APs set their α as 1.0 to maximize their throughputs. Such malicious behaviours can be resolved if other APs set their α as 1.0 as well. To investigate the effect of the malicious AP, we applied the following simulation settings in Figs. 13 and 14. First of all, three APs (i.e., AP1, AP2, and AP3) are deployed, and their SNRs are initially set to 15 dB. AP2 does not follow the strategy and fixes its channel width at 40 MHz.

In Fig. 13, AP1 and AP3 set their α as 0.2 while AP2 sets its α as 1.0. In this circumstance, AP1 and AP3 become starved because of aggressively wider channel width of AP2. On the other hand, Fig. 14 shows that AP1 and AP3 set their α

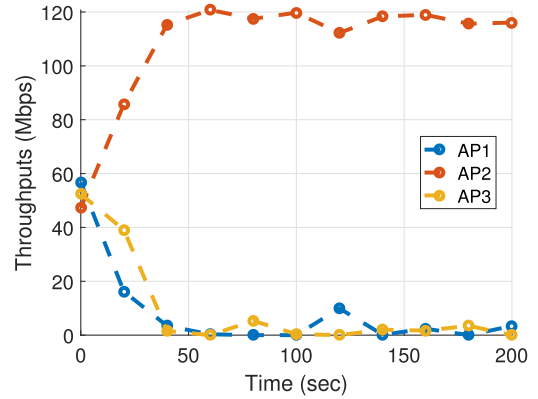
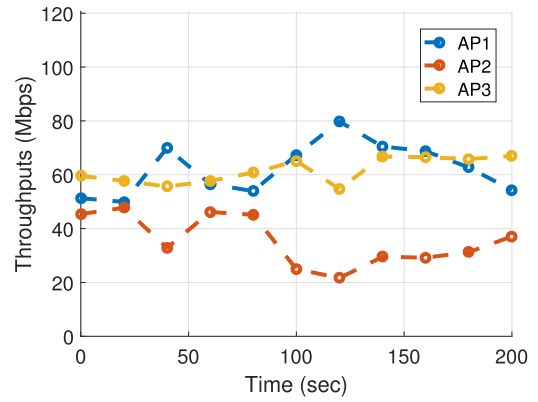
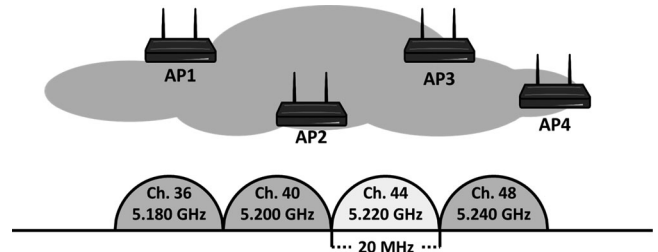
Fig. 13. Effect of malicious behaviors: $\alpha = 0.2$.Fig. 14. Effect of malicious behaviors: $\alpha = 1.0$.

Fig. 15. Network Setting.

as 1.0 as the same as AP2. Then, it can be seen that AP1 and AP3 become no longer starved and bandwidths are well-distributed among all APs.

E. Effect of Topology Change

Fig. 15 shows the effect of the topology change when an AP is newly added or an existing AP is removed. In this simulation, four APs from AP1 to AP4 are deployed, and each AP is separated by up to four channels. The SNRs of the APs are initially set to 15 dB. To simulate topology change events, AP3 is turned off at the beginning in the simulation, and AP3 is turned on at 10 s. After that, AP3 is turned off at 20 s. We let α be 0.3 in order to allow each AP to consider the adjacent APs. The mutation function is also applied so as to observe the adaptive behavior of ADRA.

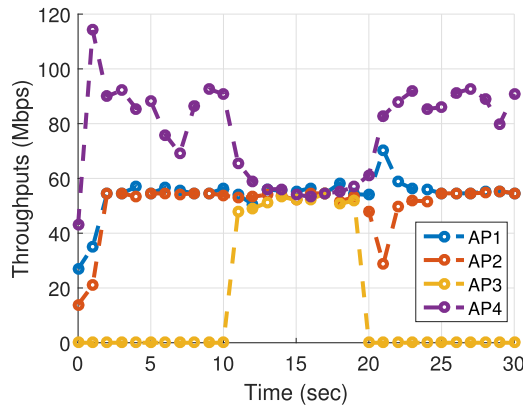


Fig. 16. Throughput variation for topology change.

In this situation, AP1 and AP2 choose the same 20 MHz and exhibit comparable throughput because they are adjacent to each other, as shown in Fig. 15. Meanwhile, AP4 can select 40 MHz and has an improved throughput because it resides far from AP1 and AP2. When AP3 is newly added at 10 s, the strategy of AP4 is affected, and thus both AP3 and AP4 set their channel widths to 20 MHz. As a result, from Fig. 16, it can be shown that AP4 experiences a reduced throughput from 10 to 20 s. On the other hand, when AP3 is removed at 20 s, AP4 senses the situation and starts to expand its channel width, which improves the throughput of AP4. The throughputs of AP1 and AP2 are rarely affected because they have already set their channel widths as 20 MHz.

VI. CONCLUSION AND FURTHER WORK

In this paper, an ADRA algorithm for densely deployed wireless LANs is proposed. By adopting a game-theoretic approach, the APs in ADRA can choose their optimal channel widths, providing a higher throughput and fair operation. The throughput and fairness of ADRA came close to those of centralized algorithms, while ADRA operates in a fully distributed manner. Extensive simulation results demonstrate that ADRA can handle dynamic environments such as a suddenly increased transmission power and maliciously calculated channel width. Furthermore, because the gain and loss functions in ADRA consider partially overlapped channels, ADRA can provide improved performance in densely deployed WLANs. In our future work, we investigate how to distribute the acquired resource to the stations, particularly in dense WLANs.

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